

WTW158 Memo - Worksheet on integration

1. $\int (\pi + \frac{1}{x}) dx = \pi x + \ln|x| + c$
2. $\int (z^4 + 4^z + e^{4z} + 4^{ez}) dz = \frac{1}{5} z^5 + \frac{1}{\ln 4} 4^z + \frac{1}{4} e^{4z} + \frac{1}{e \ln 4} 4^{ez} + c$
3. $\int x(x^2 + 1)^6 dx = \frac{1}{7} \times \frac{1}{2} (x^2 + 1)^7 + c$
4. $\int \frac{6x^2}{(x^3 + 1)^6} dx = \int 6x^2 (x^3 + 1)^{-6} dx = 6 \times \frac{1}{3} \times \frac{1}{-5} (x^3 + 1)^{-5} + c$
5. $\int (4x + 6x^2) e^{x^2+x^3} dx = 2e^{x^2+x^3} + c$
6. $\int \frac{e^{2x}}{\sqrt{1-e^{2x}}} dx = -\sqrt{1-e^{2x}} + c$
7. $\int \frac{x e^{\arctan(x^2)}}{1+x^4} dx = \frac{1}{2} e^{\arctan x^2} + c$
8. $\int \sin(30 + 4x^3) x^2 dx = -\frac{1}{12} \cos(30 + 4x^3) + c$
9. $\int x^2 \operatorname{cosec}(x^3) \cot(x^3) dx = -\frac{1}{3} \operatorname{cosec}(x^3) + c$
10. $\int \sec(\sin 2x) \tan(\sin 2x) \cos 2x dx = \frac{1}{2} \sec(\sin 2x) + c$
11. $\int \frac{\cos(\ln x)}{x} dx = \sin(\ln x) + c$
12. $\int \frac{x^9}{1+x^{10}} dx = \frac{1}{10} \ln(1+x^{10}) + c$
13. $\int \frac{x^4}{1+x^{10}} dx = \int \frac{x^4}{1+(x^5)^2} dx = \frac{1}{5} \arctan x^5 + c$
14. $\int \frac{1}{(1+4x^2) \arctan 2x} dx = \frac{1}{2} \ln(\arctan 2x) + c$
15. $\int \frac{x^2 \arcsin x^3}{\sqrt{1-x^6}} dx = \frac{1}{2} \times \frac{1}{3} (\arcsin x^3)^2 + c$
16. $\int \frac{x^3}{\sqrt{1-x^8}} dx = \int \frac{x^3}{\sqrt{1-(x^4)^2}} dx = \frac{1}{4} \arcsin x^4 + c$
17. $\int \frac{x^7}{\sqrt{1-x^8}} dx = \int x^7 (1-x^8)^{-1/2} dx = -2 \times \frac{1}{8} \sqrt{1-x^8} + c$
18. $\int \frac{e^{-x}}{\sqrt{1-e^{-2x}}} dx = \int \frac{e^{-x}}{\sqrt{1-(e^{-x})^2}} dx = -\arcsin(e^{-x}) + c$
19. $\int 6^{2t} dt = \frac{1}{2 \ln 6} 6^{2t} + c$
20. $\int 3^y \sqrt{1-3^y} dy = -\frac{2}{3 \ln 3} (1-3^y)^{3/2} + c$

DEFINITE INTEGRALS AND AREAS

Evaluate the following definite integrals.

$$1. \int_{-2}^{-1} (x-5) dx = \left(\frac{1}{2}x^2 - 5x \right) \Big|_{-2}^{-1} = -\frac{13}{2}$$

$$2. \int_{-2}^{-1} \frac{1}{x} dx = \ln|x| \Big|_{-2}^{-1} = -\ln 2$$

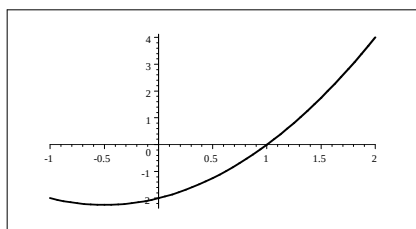
$$3. \int_{-2}^{-1} \frac{1}{x^2} dx = \int_{-2}^{-1} x^{-2} dx = -x^{-1} \Big|_{-2}^{-1} = -(-1 + \frac{1}{2}) = \frac{1}{2}$$

$$4. \int_0^{\frac{\pi}{2}} (\sin x - 4 \cos 3x) dx = (-\cos x - \frac{4}{3} \sin 3x) \Big|_0^{\pi/2} = \frac{7}{3}$$

$$5. \int_{-2}^{-1} \frac{x}{9+x^2} dx = \frac{1}{2} \ln|9+x^2| \Big|_{-2}^{-1} = \frac{1}{2} \ln 10 - \frac{1}{2} \ln 13$$

$$6. \int_0^1 \frac{1}{9+x^2} dx = \int_0^1 \frac{1}{9(1+(x/3)^2)} dx = \frac{1}{9} \times 3 \arctan(x/3) \Big|_0^1 \\ = 3 \arctan \frac{1}{3} - 3 \arctan 0 = 3 \arctan \frac{1}{3}$$

7.1



$$7.2 \int_{-1}^1 (x^2 + x - 2) dx = -\frac{10}{3} \text{ and } \int_1^2 (x^2 + x - 2) dx = \frac{11}{6}$$

$$\text{The area is } \frac{10}{3} + \frac{11}{6} = \frac{31}{6}$$

$$8.1 \int f(x) dx = \int x e^{x^2} dx = \frac{1}{2} e^{x^2} + c$$

$$8.2 \int_{-1}^1 f(x) dx = \int_{-1}^1 x e^{x^2} dx = 0$$

$$8.3 f(-x) = (-x)e^{(-x)^2} = -x e^{x^2} = -f(x).$$

$$8.4 \text{ As } f \text{ is an odd function and } f(x) > 0 \text{ if } x > 0 \text{ it follows that } \int_{-a}^0 f(x) dx = -\int_0^a f(x) dx$$

$$\text{The area is } 2 \int_0^1 f(x) dx = 2 \left(\frac{1}{2} e - \frac{1}{2} \right) = e - 1$$

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